

AE 413 – Final Project

Aircraft Design, Simulation and Analysis

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EMBRY-RIDDLE AERONAUTICAL UNIVERSITY

Daytona Beach Campus

AE 413: Aircraft Stability and Control

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Abstract

This project presents the design, aerodynamic, and stability evaluation of a proposed aircraft capable of meeting longitudinal, directional, and lateral stability requirements. The aircraft is designed to fly at Mach 0.6 and 6,500 ft, with a weight of 4,000 lb and is optimized for a flight condition at 2° angle of attack, although the DATCOM could determine stability and control coefficients for more than 3 different flight conditions. The initial geometry for the aircraft was performed manually and resulted in a wingspan 22 ft, wing area 121 ft², fuselage length of 27.7ft and an aspect ratio of 4.0. The main aerodynamic chord fit in the range of requirements, as well as the sweep and dihedral angles for the wings, which resulted in 11 degrees and 3 degrees respectively. The resulting aerodynamic coefficients were $C_{L_0} = 0.143$, $C_{D_0} = 0.017$, $C_{L_\alpha} = 4.648 \text{ rad}^{-1}$, $C_{m_\alpha} = -0.293 \text{ rad}^{-1}$, and $C_{n_\beta} = 0.183 \text{ rad}^{-1}$, which also meet all of the stated requirements.

Stability derivatives were calculated through both analytical methods and DATCOM. Results for C_{n_β} and C_{l_β} showed similar values between the two approaches, with values of 0.183 rad^{-1} and -0.113 rad^{-1} from DATCOM, which closely match the manual results of 0.189 rad^{-1} and -0.123 rad^{-1} , respectively. However, there is a significant discrepancy in the C_{m_α} calculated through DATCOM and manually. For DATCOM, the value was -0.293 rad^{-1} , while for the hand calculation it resulted in a value -1.35 rad^{-1} , this emphasizes the limitations of DATCOM's modeling for certain aircraft geometries.

To validate the design's dynamic behavior, a 3D model was done in MATLAB, and also integrated into Simulink, and simulated flying conditions using FlightGear. The aircraft's stability and control responses, which were calculated and provided through DATCOM, such as $C_{m_Q} = -0.1992$ and $C_{n_r} = -0.00843$ demonstrate a satisfactory performance under various maneuvers, including level flight, pitch-up. Moreover, the agreement from both analytical hand calculating processes and DATCOM results supports most of the findings, as well as reinforces a correct simulation and the proposed aircraft design's compliance with stability and control criteria.

Introduction

This report examines the longitudinal, directional, and lateral stability and controllability of an aircraft designed to meet the specific requirements established by our design group. The configuration adheres to several geometric constraints, including wingspan, wing surface area, and sweep angle, among others. Additionally, the design satisfies key aerodynamic stability parameters such as target values for C_{L_0} , C_{D_0} , C_{L_α} , C_{m_α} , and C_{n_β} . The proposed aircraft weighs 4,000 lb, and can be at or around trimmed level flight at an altitude of approximately 6,500 feet and a subsonic speed of Mach 0.6.

The design process began with hand calculation of the aircraft's geometric parameters to ensure compliance with the requirements. All these constraints, then, had to be verified for a specific

flight condition. The selected configuration for the proposed aircraft is for a flight condition of an angle of attack (α) of 2 degrees.

The use of DATCOM was necessary to accurately determine the aircraft's stability derivatives and control responses. It helped to compute aerodynamic coefficients and stability derivatives over multiple flight conditions. Then we integrated into Simulink five of the derivatives. Additionally, we used a MATLAB code provided by the instructor to create a 3D model of our aircraft with all the selected characteristics. Additionally, the model was then run in the FlightGear interface to visualize how the aircraft performs under a simulation.

This report is an extensive summary of a full design cycle of stability and control of an aircraft: from initial geometry estimation through hand calculations to software-based aerodynamic analysis and flight simulation. Various tools were used in order to obtain accurate results and verify the results. Moreover, an in-depth analysis and comparison of the hand calculations and DATCOM is provided to represent the involvement and effort of the project.

Aircraft Geometry and Properties

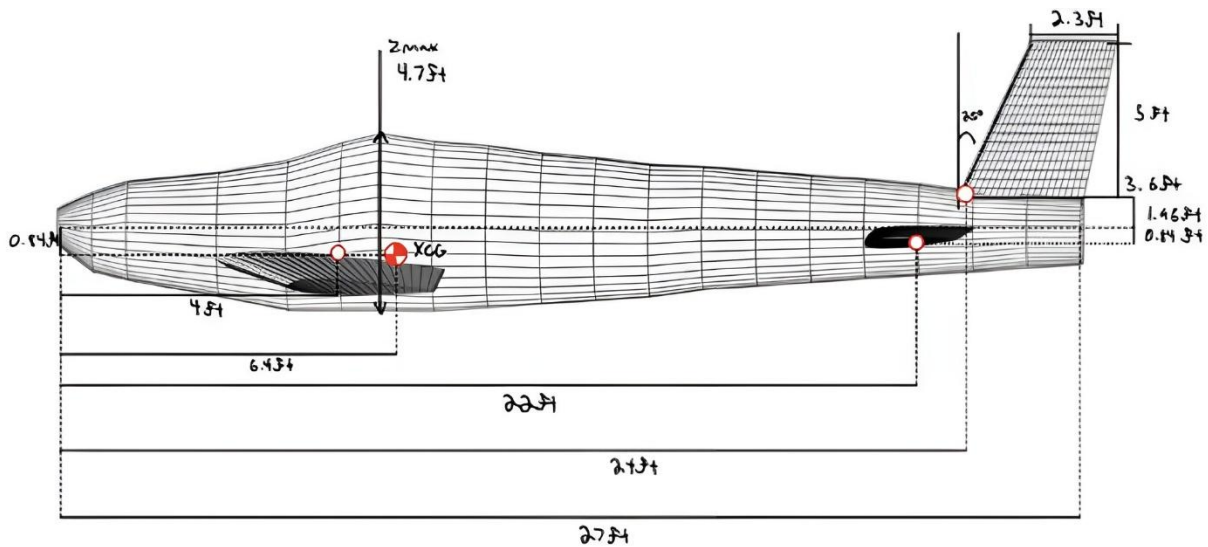
This section contains tables displaying all of the proposed aircraft geometry and properties. Additionally, a diagram displaying the top, lateral, and back views with all dimensions and geometry.

Parameters (Wing)	Dimension
Root Chord	6.5 ft
Tip Chord	4.5 ft
Wingspan	22 ft
Taper Ratio	0.692
Wing Surface Area	121 ft ²
Mean Aerodynamic Chord	5.561 ft
Aspect Ratio	4
Sweep Angle	11 deg
Dihedral Angle	3 deg

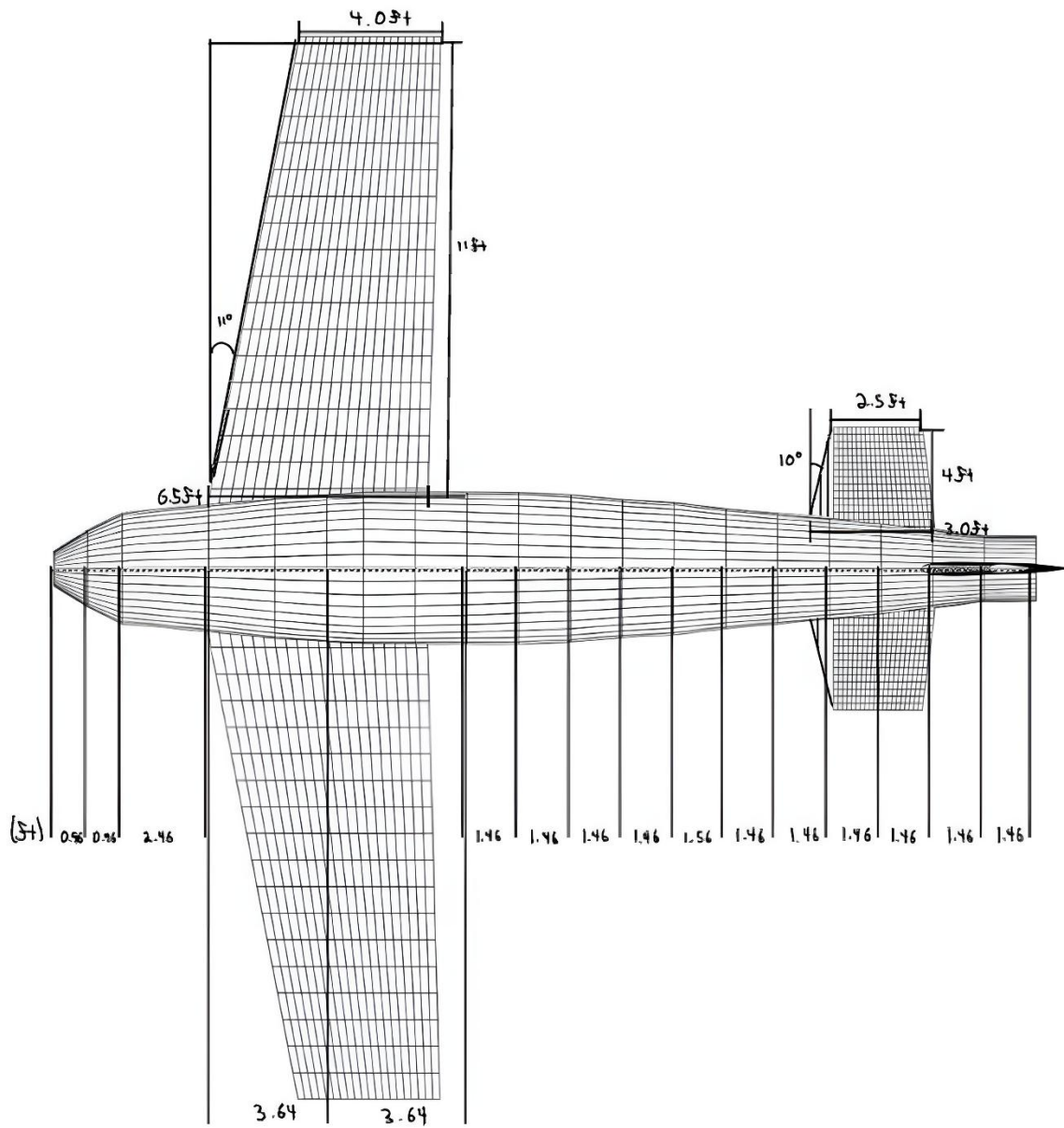
Parameters (Horizontal Tail)	Dimension
Root Chord	3 ft
Tip Chord	2.5 ft
Wingspan	8 ft
Taper Ratio	0.833
Wing Surface Area	22 ft ²
Mean Aerodynamic Chord	2.758 ft
Aspect Ratio	2.909
Sweep Angle	10 deg

Dihedral Angle	8 deg
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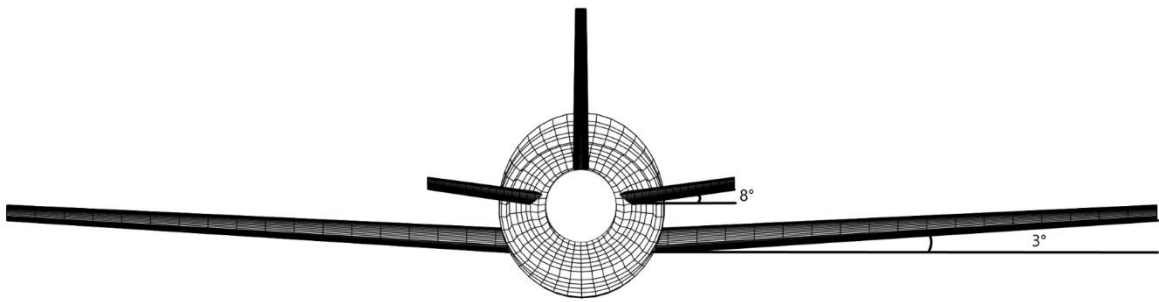
Parameters	Dimension
Angle of Attack	2 deg
Weight	4000 lb
Length of Fuselage	27.708 ft
Altitude	6500 ft
Mach	0.6
Total mass	4000 lb
Passengers	2
C_{L_0}	0.143
C_{D_0}	0.017
C_{L_α}	4.648 per rad
C_{m_α}	-0.293 per rad
C_{n_β}	0.183 per rad



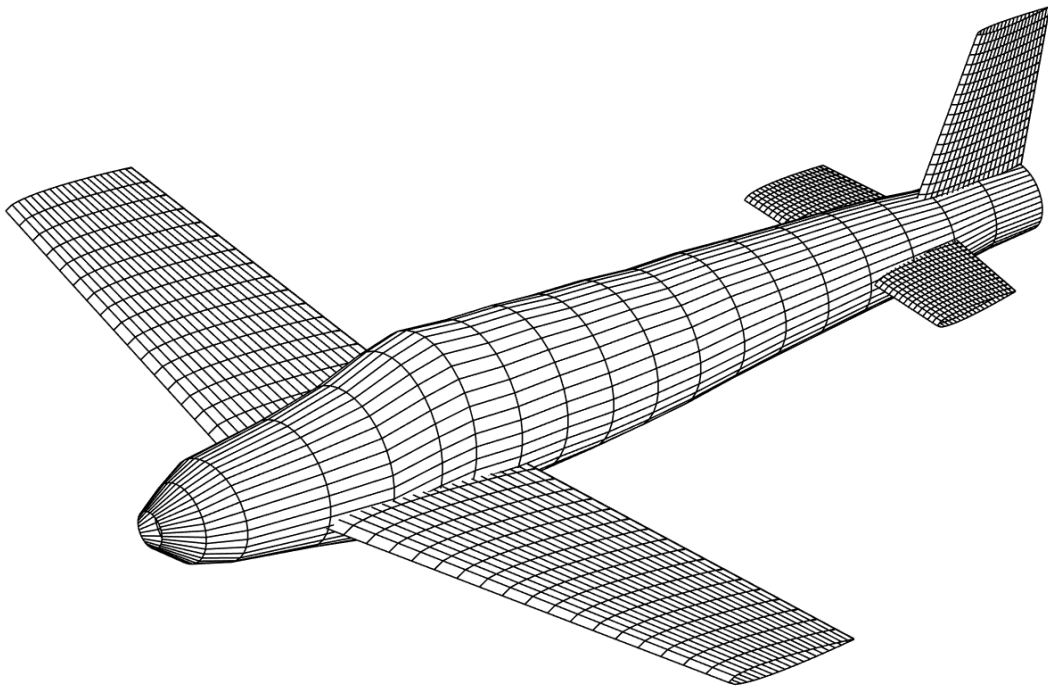
CAD Side View of Aircraft



CAD Top View of Aircraft



CAD Back View of Aircraft



CAD Orthogonal View of Aircraft

Longitudinal Stability Analysis (Hand calculations, DATCOM)

Hand Calculations:

$$\begin{aligned}\frac{\partial \epsilon}{\partial \alpha} &= 4.44 \left[K_A K_\lambda K_H (\cos \Lambda_{c/H})^{1.19} \right] \\ K_A &= \frac{1}{A} - \frac{1}{1+A^{1.25}}, \quad K_\lambda = \frac{10-3\lambda}{7}, \quad K_H = \frac{1-h_H/b}{\sqrt[3]{2} h_H/b}, \quad \tan \Lambda_{c/H} = \tan \Lambda_{c/E} - \frac{4(0.25)(1-\lambda)}{A(1+\lambda)} \\ K_A &= \frac{1}{4} - \frac{1}{1+4^{1.25}} = 0.1634 \\ K_\lambda &= \frac{10-3(0.6923)}{7} = 1.132 \\ K_H &= \frac{1-(1.967/22)}{\sqrt[3]{2} (16.043/22)} = 0.8025 \\ \tan \Lambda_{c/H} &= \tan(11) - \frac{4(0.25)(1-0.6923)}{4(1+0.6923)} = 0.1489 \\ \Lambda_{c/H} &= \tan^{-1}(0.1489) = 8.469 \\ \frac{\partial \epsilon}{\partial \alpha} &= 4.44 \left[0.1634 (1.132) (0.8025) (\cos(8.469))^{1.19} \right] = 0.456\end{aligned}$$

$$\begin{aligned}C_{m_\alpha} &= C_{m_\alpha}_F + C_{m_\alpha}_w - C_{m_\alpha}_{ht} \\ C_{m_\alpha}_F &\text{ is given by Multhopp code to be } \Rightarrow 0.4702 \\ C_{m_\alpha}_w &= C_{L_\alpha_w} \bar{x}_a = 4.247(-0.07) = -0.297 \\ C_{m_\alpha}_{ht} &= C_{L_{ht}} \eta \bar{v} \left(1 - \frac{\partial \epsilon}{\partial \alpha}\right) = 4.71(1)(0.6) (1 - 0.456) = 1.53 \\ C_{m_\alpha} &= 0.4702 - 0.297 - 1.53 = -1.3575\end{aligned}$$

See appendix II) a. Multhopp Script by Meagan Shivers, for $C_{m_{\alpha,f}}$

Datcom Values:

FLIGHT CONDITIONS						REFERENCE DIMENSIONS				
MACH NUMBER	ALTITUDE	VELOCITY	PRESSURE	TEMPERATURE	REYNOLDS NUMBER	REF. AREA	REFERENCE LENGTH	MOMENT	REF. CENTER	
	FT	FT/SEC	LB/FT**2	DEG R	1/FT	FT**2	FT	FT	FT	FT
0 0.600	6500.00	654.66	1.6643E+03	495.497	3.5364E+06	121.000	5.560	22.000	6.400	-0.840
0 0										
-----DERIVATIVE (PER DEGREE)-----										
0 ALPHA	CD	CL	CM	CN	CA	XCP	CLA	CMA	CYB	CNB
0										CLB
-4.0	0.019	-0.168	0.1055	-0.169	0.008	-0.623	7.648E-02	-1.508E-02	-Inf	3.193E-03
-2.0	0.016	-0.014	0.0793	-0.014	0.016	-5.481	7.772E-02	-1.055E-02		-1.371E-03
0.0	0.017	0.143	0.0633	0.143	0.017	0.444	7.915E-02	-5.842E-03		-1.520E-03
2.0	0.023	0.303	0.0560	0.303	0.012	0.185	8.109E-02	-5.119E-03		-1.674E-03
4.0	0.033	0.467	0.0428	0.468	0.000	0.091	8.277E-02	-5.526E-03		-1.980E-03
6.0	0.048	0.634	0.0339	0.635	-0.019	0.053	8.216E-02	-4.357E-03		-2.002E-03
8.0	0.067	0.796	0.0254	0.797	-0.044	0.032	7.393E-02	-1.714E-02		-2.173E-03
10.0	0.087	0.930	-0.0347	0.930	-0.076	-0.037	6.151E-02	-3.272E-02		-2.331E-03
12.0	0.106	1.042	-0.1055	1.041	-0.113	-0.101	5.052E-02	-3.947E-02		-2.430E-03
14.0	0.126	1.132	-0.1926	1.128	-0.152	-0.171	4.045E-02	-4.296E-02		-2.479E-03
16.0	0.146	1.203	-0.2774	1.197	-0.192	-0.232	6.726E-03	-4.447E-02		-2.469E-03
18.0	0.150	1.158	-0.3705	1.148	-0.216	-0.323	-5.164E-02	-4.862E-02		-2.414E-03
										-2.135E-03

$$C_{m_{\alpha}} = -0.005119 / \text{deg} = -0.2943 / \text{rad}$$

In order to manually calculate the value for $C_{m_{\alpha}}$, class notes from 413 Stability and Control class were used. This process ensured our value aligned with typical in class results, suggesting the aircraft's static longitudinal stability values account for all variables and a correct value is achieved. However, the DATCOM for $C_{m_{\alpha}}$, significantly differs from the hand calculated result. This discrepancy in the result may arise due to limitations in the DATCOM modeling assumptions, which can yield less accurate results for certain configurations or geometries. This reinforces the need to cross verify DATCOM outputs with analytical methods to ensure values are accurate.

Directional Stability Analysis (Hand calculations, DATCOM)

Hand Calculations:

(5pts) Wing sweep contribution to $C_{n\beta}$
 (5pts) Vertical tail contribution to $C_{n\beta}$
 (2pts) Total $C_{n\beta}$

$$C_{n\beta}_w = -K_{N, \alpha_{L1}} \frac{S_{w, L1}}{S_w b_w} = -0.0021 (1.95) \left(\frac{84 \cdot 27.709}{121 \cdot 22} \right) = -3.58 \times 10^{-3} / \text{deg or } -0.2052 / \text{rad}$$

$$R_{\text{ymils}} = \frac{654.66 (27.7)}{1.79 \times 10^4} = 1.01 \times 10^3 \Rightarrow K_{N1} = 1.95$$

$$K_{N1} = 0.0021$$

$$C_{n\beta}_{r,w} = -\frac{2r}{S_w b_w} \int_0^{b/2} (C_L - C_{D\alpha}) C(y) y dy$$

$$= -\frac{2(3\pi)}{121 \cdot 22} \int_0^{22/2} (0.291 - 0.204) (6.5 - 1.18/y) y dy = -0.00805 / \text{rad}$$

$$C_{n\beta}_{A,w} = \frac{2 \sin \Lambda}{S_w b_w} \int_0^{b/2 \sec \Lambda} [C_{L_{\alpha}}(\alpha)^2 \sec \Lambda + 2 C_{D_{\alpha}} \sec \Lambda + C_{D_{\alpha}} \alpha] C(y_n) y_n dy_n$$

$$= \frac{2 \sin 3}{121 \cdot 22} \int_0^{11.01} [(4.247)(0.035)(1000) + 2(0.017)(0.999) + 0.00355(2)] (6.49) y_n dy_n = 0.000715 / \text{rad}$$

$$C_{n\beta}_{v,b} = K_{A,v} C_{L_{\alpha,v}} n_v \left(1 + \frac{\partial \sigma}{\partial \beta} \right) \Rightarrow n_v \left(1 + \frac{\partial \sigma}{\partial \beta} \right) = 0.724 + \frac{3.06}{1 + \cos(8.469)} \frac{19.08}{121} + \frac{0.44}{2(2.156)} + 0.009(4) = 1.3736$$

$$(0.069)(4.247)(1.3736) = 0.4025 / \text{rad}$$

$$C_{n\beta}_{\text{total}} = -0.2052 - 0.00805 + 0.000715 + 0.4025 = 0.189 / \text{rad}$$

Datcom Values:

FLIGHT CONDITIONS						REFERENCE DIMENSIONS					
MACH NUMBER	ALTITUDE	VELOCITY	PRESSURE	TEMPERATURE	REYNOLDS NUMBER	REF. AREA	REFERENCE LENGTH	MOMENT	REF. CENTER		
	FT	FT/SEC	LB/FT**2	DEG R	1/FT	FT**2	LONG. FT	LAT. FT	HORIZ FT	VERT FT	
0 0.600	6500.00	654.66	1.6643E+03	495.497	3.5364E+06	121.000	5.560	22.000	6.400	-0.840	
0	-----DERIVATIVE (PER DEGREE)-----										
0 ALPHA	CD	CL	CM	CN	CA	XCP	CLA	CMA	CYB	CLB	
0											
-4.0	0.019	-0.168	0.1055	-0.169	0.008	-0.623	7.648E-02	-1.508E-02	-Inf	3.193E-03	
-2.0	0.016	-0.014	0.0793	-0.014	0.016	-5.481	7.772E-02	-1.055E-02		-1.520E-03	
0.0	0.017	0.143	0.0633	0.143	0.017	0.444	7.915E-02	-5.842E-03		-1.674E-03	
2.0	0.023	0.303	0.0560	0.303	0.012	0.185	8.109E-02	-5.119E-03		-1.980E-03	
4.0	0.033	0.467	0.0428	0.468	0.000	0.091	8.277E-02	-5.526E-03		-2.002E-03	
6.0	0.048	0.634	0.0339	0.635	-0.019	0.053	8.216E-02	-4.357E-03		-2.173E-03	
8.0	0.067	0.796	0.0254	0.797	-0.044	0.032	7.393E-02	-1.714E-02		-2.331E-03	
10.0	0.087	0.930	-0.0347	0.930	-0.076	-0.037	6.151E-02	-3.272E-02		-2.430E-03	
12.0	0.106	1.042	-0.1055	1.041	-0.113	-0.101	5.052E-02	-3.947E-02		-2.479E-03	
14.0	0.126	1.132	-0.1926	1.128	-0.152	-0.171	4.045E-02	-4.296E-02		-2.469E-03	
16.0	0.146	1.203	-0.2774	1.197	-0.192	-0.232	6.726E-03	-4.447E-02		-2.414E-03	
18.0	0.150	1.158	-0.3705	1.148	-0.216	-0.323	-5.164E-02	-4.862E-02		-2.135E-03	

$$C_{n\beta} = 0.003193 / \text{deg} = 0.1829 / \text{rad}$$

The DATCOM result for $C_{n\beta}$ is really close to our hand calculated result, the percentage error between these two values is of less than 3.5%. This indicates that for this case DATCOM provides a reliable estimate for the directional stability as well as provides consistency with hand calculations. Hence, supporting the validity of both methods for evaluating lateral-directional stability.

Lateral Stability Analysis (Hand calculations, DATCOM)

Hand Calculations:

$$\begin{aligned}
 C_{l\beta} \text{ TOTAL} &= C_{l\beta} \text{ A/R} + C_{l\beta} \text{ V} \\
 C_{l\beta} \text{ A/R} &= - \frac{2(\Gamma + \alpha \tan \Lambda)}{S_w b_w} \int_0^{b_w/2} a_o(y) c(y) y \, dy \\
 &= - \frac{2(0.05236 + 0.00676)}{121 \cdot 22} \int_0^{22/2} (4.24)(6.5 - 0.183 y) y \, dy = \\
 &= \boxed{-0.05874 / \text{rad}} \\
 C_{l\beta} \text{ V} &= C_{y\beta} \left(\frac{z}{b_w} \right) \Rightarrow C_{y\beta} = -k C_{L\alpha} \left(1 + \frac{\partial \sigma}{\partial \beta} \right) n_v \left(\frac{S_v}{S_w} \right) = -1.398 / \text{rad} \\
 &= -1.398 \left(\frac{1.013}{22} \right) = \boxed{-0.06437 / \text{rad}} \\
 C_{l\beta} \text{ TOTAL} &= (-0.05874) + (-0.06437) = \boxed{-0.1231 / \text{rad}}
 \end{aligned}$$

Datcom Values:

FLIGHT CONDITIONS				REFERENCE DIMENSIONS			
MACH NUMBER	ALTITUDE	VELOCITY	PRESSURE	TEMPERATURE	REYNOLDS NUMBER	REF. AREA	REFERENCE LENGTH
0 0.600	FT	FT/SEC	LB/FT**2	DEG R	1/FT	FT**2	FT
0	6500.00	654.66	1.6643E+03	495.497	3.5364E+06	121.000	5.560
0 ALPHA	CD	CL	CM	CN	CA	XCP	CLA
0	0.019	-0.168	0.1055	-0.169	0.008	-0.623	7.648E-02
-4.0	0.016	-0.014	0.0793	-0.014	0.016	-5.481	7.772E-02
-2.0	0.017	0.143	0.0633	0.143	0.017	0.444	7.915E-02
0.0	0.023	0.303	0.0560	0.303	0.012	0.185	8.109E-02
2.0	0.033	0.467	0.0428	0.468	0.000	0.091	8.277E-02
4.0	0.048	0.634	0.0339	0.635	-0.019	0.053	8.216E-02
6.0	0.067	0.796	0.0254	0.797	-0.044	0.032	7.393E-02
8.0	0.087	0.930	-0.0347	0.930	-0.076	-0.037	6.151E-02
10.0	0.106	1.042	-0.1055	1.041	-0.113	-0.101	5.052E-02
12.0	0.126	1.132	-0.1926	1.128	-0.152	-0.171	4.045E-02
14.0	0.146	1.203	-0.2774	1.197	-0.192	-0.232	6.726E-03
16.0	0.150	1.158	-0.3705	1.148	-0.216	-0.323	-5.164E-02
18.0							

$$C_{l\beta} = -0.00198 / \text{deg} = -0.113 / \text{rad}$$

Our DATCOM value for the total $C_{l\beta}$ is really close to our hand calculated result. The difference is just of 0.01/rad; therefore, showing both results fall within a reasonable range and consistent sign and behavior. This indicates good agreement between DATCOM and analytical methods, suggesting that DATCOM provided a reliable estimate of lateral stability for this configuration.

Comparison table between hand calculations and DATCOM for one specific flight condition

Angle of attack = 2 deg.

Some parameters are not produced by DATCOM especially longitudinal, directional, and lateral control with neutral point. Everything else is mostly provided and both hand-calculated and DATCOM values match except some for which analysis already provided in earlier sections of the report.

Parameters	Hand-Calculated	DATCOM
$C_{L\alpha}$	0.0741 /deg	0.08109 /deg
$C_{D\alpha}$	0.0035 /deg	N/A
C_L	0.2910	0.303
C_D	0.024	0.023
V_{stall}	163.8 ft/sec	163.8 ft/sec
L	68785.07 N	68785.075 N
D	5693.279 N	5693.279 N
$\frac{d\epsilon}{d\alpha}$	0.456	0.461
$C_{m\alpha}$	-1.35 /rad	-0.293 /rad
$C_{n\beta}$	0.189 /rad	0.183 / rad
$C_{l\beta}$	-0.1231 /rad	-0.113 / rad
$C_{l\delta,a}$	0.0286 /rad	N/A
$C_{n\delta,r}$	-0.0075 /deg	N/A
$C_{m\delta,e}$	-0.01385 /deg	N/A
N_0	0.435/43.5 %	N/A

Hand Calculations

$$1) C_{L\alpha} = \frac{2\pi AR}{2 + \sqrt{4 + \left(\frac{AR}{\gamma}\right)^2}} = \frac{2\pi(4)}{2 + \sqrt{4 + \left(\frac{4(\sqrt{1-0.62})}{0.95}\right)^2}} = \frac{4.247}{\text{rad}} \quad \boxed{0.0741 \text{ deg}}$$

$$2) C_{D\alpha} = \frac{C_D - C_{D0}}{\alpha} = \frac{0.024 - 0.017}{2} = 3.5 \times 10^{-3}$$

$$3) C_L = C_{L\alpha} \cdot \alpha + C_{L0}$$

$$C_L = 0.0741(2) + 0.143 = \boxed{0.291}$$

$$4) C_D = C_{D0} + K C_L^2$$

$$C_D = 0.017 + (0.0837)(0.2912)^2 = \boxed{0.024}$$

$$5) V_{\text{stall}} = \sqrt{\frac{2W}{\rho S C_{L_{\text{max}}}}} = \sqrt{\frac{2(4000)}{(0.002049)(121)(1.203)}} = \boxed{163.8 \text{ ft/s}}$$

$$6) L = C_L \frac{1}{2} \rho V^2 S = (0.291) \left(\frac{1}{2}\right) (0.002049) (654.66)^2 (121) = 15463.5 \text{ lb}$$

$$\boxed{68785.07 \text{ N}}$$

$$7) D = C_D \frac{1}{2} \rho V^2 S = (0.024) \left(\frac{1}{2}\right) (0.002049) (654.66)^2 (121) = 1279.9 \text{ lb}$$

$$\boxed{5693.2 \text{ N}}$$

(10pts) Clearly present the calculation of fuselage contribution, wing contribution, vertical tail contribution, total $C_{m\delta e}$, $C_{n\delta r}$, $C_{l\delta a}$

$$C_{l\delta a} = \frac{2 \cdot C_{L\alpha} \cdot \tau_a}{S \cdot b} \int_{y_1}^{y_2} C_a(y) y dy = \frac{2 \cdot C_{L\alpha} \cdot \tau_a}{S \cdot b} \int_{y_1}^{y_2} C_r \left[1 + \left(\frac{\lambda-1}{b/2} \right) y \right] y dy$$

$$= \frac{2 \cdot (4.2411) \cdot (0.45)}{121 \cdot 22} \int_{6.58}^{11} 6.5 \left[1 + \frac{(0.692-1)}{(11)} y \right] y dy = 0.0286 / \text{rad}$$

$$C_{n\delta r} = -k_{n_v} \bar{V}_2 C_{L\alpha_v} \tau_e = -(0.9)(0.25)(0.0741)(0.45) = -7.502 \times 10^{-3} / \text{deg}$$

$$C_{m\delta e} = -V_e \eta C_{L\alpha_e} \tau_e = (0.9)(0.4)(0.074)(0.52) = -0.01385 / \text{deg}$$

(5pts) The Neutral point N of an aircraft determines the most aft center of gravity (CG) limit in a stable aircraft. If you place the CG behind (aft) of the neutral point, you will get an unstable airplane. Find the neutral point of your proposed airplane.

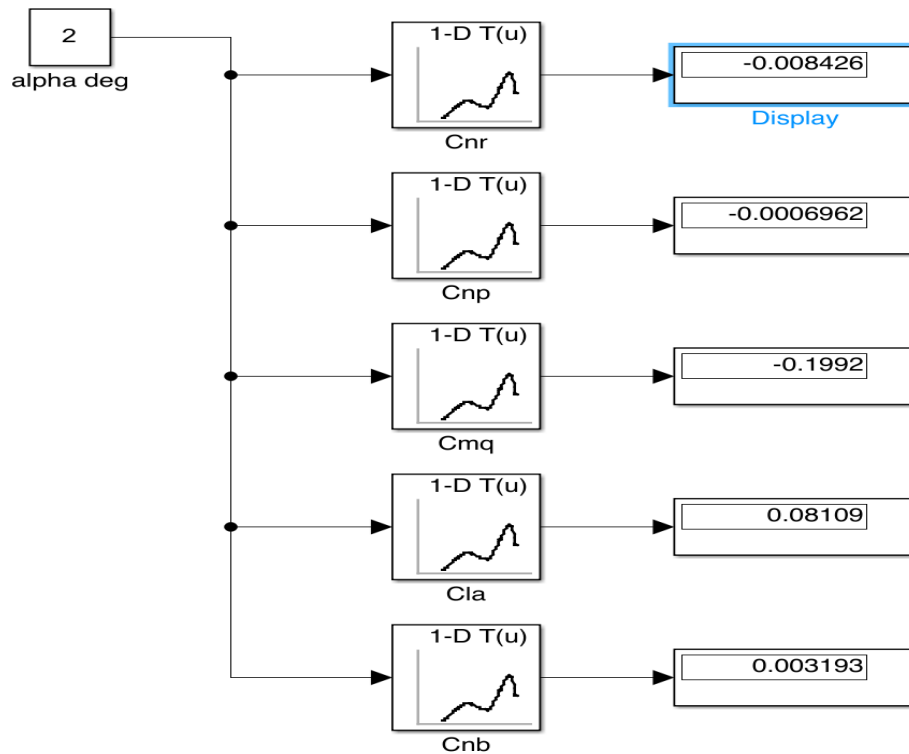
$$N_0 = \bar{x}_{cg} - \left(\frac{\partial C_m}{\partial C_L} \right)_{\text{Fixed Stick}} = 0.25 - \left(\frac{0 - 0.056}{0.303 - 0} \right) = 0.435 \text{ or } 43.5\%$$

Summary of Look-up Tables

Below is a table showing the summary of look-up values. These values were directly obtained from DATCOM and compared to the look-up table reference provided from a MATLAB zip code that the instructor shared. These values were then integrated into Simulink.

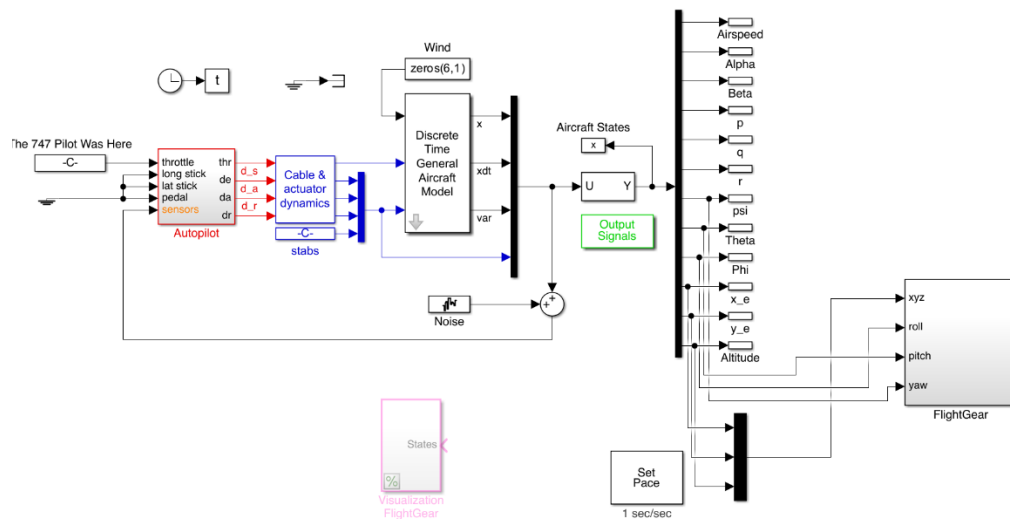
Parameters	Simulink Integration
C_{nr}	-0.008426
C_{np}	-0.0006962
C_{mq}	-0.1992
C_{la}	0.08109
C_{nb}	0.003193

The following image displays the Simulink setup:



Simulink Simulation Environment (Aerodynamics, FlightGear)

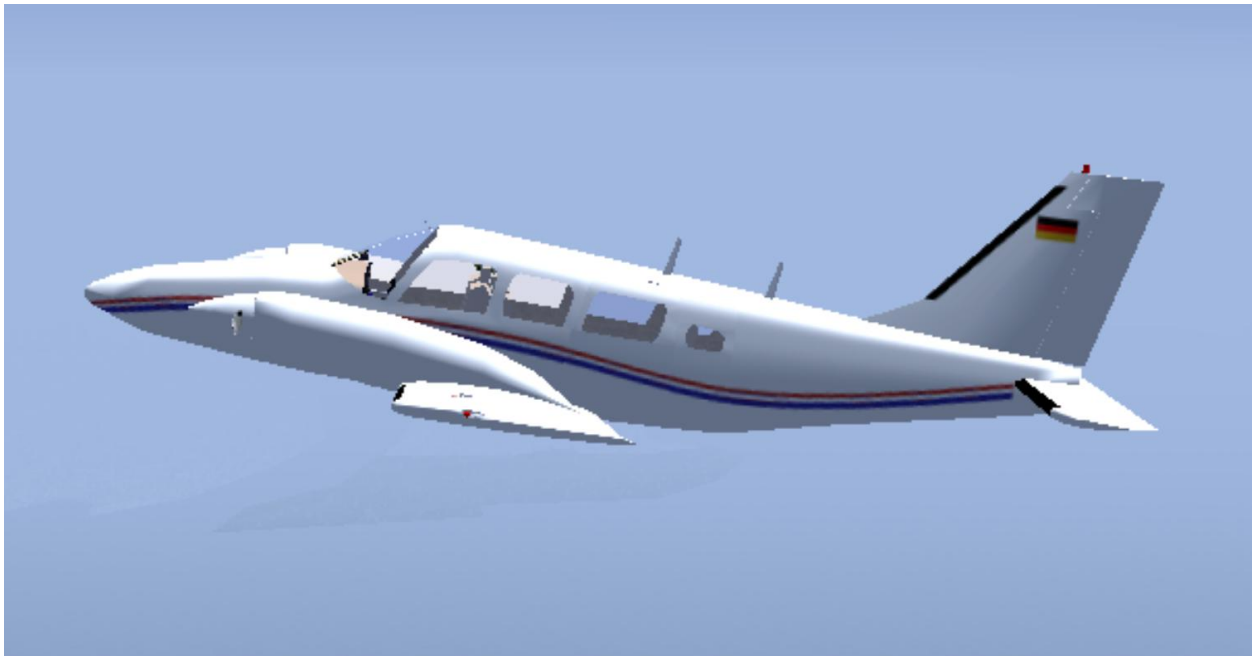
The figure below depicts the FlightGear interface using Simulink for our proposed aircraft.



Simulation Results:



Level flight: ($V = 654\text{ft/s}$, Aileron= 0 deg, Rudder = 0 deg, Elevator = -1.71deg, Alt = 6500ft)



Pitch Up: ($V = 630\text{ft/s}$, Aileron= 0 deg, Rudder = 0 deg, Elevator = 2deg, Alt = 6530ft)



Right Bank: ($V = 654\text{ft/s}$, Aileron = 0 deg, Rudder = 0 deg, Elevator = 0deg, Alt = 6500ft)

Conclusion

This project demonstrated the full cycle of aircraft stability and control analysis, from initial hand calculations to software-based aerodynamic modeling and flight simulation. Key aerodynamic derivatives such as C_{L_α} , C_{m_α} , and C_{n_β} were calculated both analytically and using USAF DATCOM. The DATCOM output for C_{n_β} and C_{l_β} closely matched hand calculations; however, the significant difference observed for the C_{m_α} value highlights the limitations of DATCOM for certain aircraft configurations. Thus, it emphasizes the importance of validating computational results with analytical methods. In conclusion, the aircraft meets the stability requirements and shows consistency between most of the DATCOM and hand-calculated values, confirming the design's validity. The integration of DATCOM outputs into Simulink and visualization through FlightGear further depicts the analysis.

References

1. *DATCOM import file.* USAF. Aerospace Engineering Department, Embry-Riddle Aeronautical University.

2. Greiner, G. P., & Mohammed, J. (2008). *datcom3d input file*. Aerospace Engineering Department, Embry-Riddle Aeronautical University.
3. Shivers, M. (n.d.). *Multhopp's Theory Script*. Embry-Riddle Aeronautical University.

Appendix (DATCOM, SIMULINK codes)

I)

a. DATCOM input file:

```
CASEID AE413 TEST
$FLTCN NMACH=1.0,MACH(1)=0.6$
$FLTCN NALT=1.0,ALT(1)=6500.0$
$FLTCN NALPHA=12.0,ALSCHD=-4.0,-2.0,0.0,2.0,4.0,6.0,8.0,
10.0,12.0,14.0,16.0,18.0,GAMMA=0.0,WT=4000.0,LOOP=1.0$
$OPTINS SREF=121.0,CBARR=5.56,BLREF=22.0$
$SYNTHS XCG=6.4,ZCG=-0.84,XW=4.0,ZW=-0.84,ALIW=2.0,XH=22.0,
ZH=0.0,ALIH=-3.0,XV=24.0,ZV=0.125,XVF=19.76,ZVF=1.25,VERTUP=.TRUE.$
$BODY NX=20.0,ITYPE=1.0,
X=0.0,0.958,1.9165,4.375,6.25,7.708,8.75,10.208,11.6665,13.125,14.583,
16.0415,17.5,18.958,20.41665,21.875,23.333,24.79165,26.25,27.708,
ZU=0.0,0.8835,1.1835,1.43750,1.76875,2.26040,2.5,2.2915,2.1875,1.9875,
1.8771,1.70625,1.659585,1.54854,1.4354,1.3415,1.209375,1.0098335,0.98229,0.87125,
ZL=-0.1,-1.271,-1.4875,-1.9146,-2.296,-2.296,-2.2915,-2.2915,-2.1875,-2.0835,
-1.9790,-1.875,-1.6665,-1.56250,-1.4585,-1.3540,-1.25,-1.146,-1.0415,-0.9375,
S=0.0,0.9765,1.177,1.3735,2.3145,2.2915,2.196,2.1005,2.005,1.8758,
1.65625,1.453605,1.38605,1.31720,1.24126,1.096,1.08434,0.853125,0.7520,0.51215,
P=6.913,8.999,9.668,11.207,12.683,13.176,14.114,15.019,15.399,
15.533,15.003,14.765,13.749,11.702,9.299,7.039,5.618,2.292,
R=0.0,1.090,1.568,1.803,1.999,2.097,2.156,2.156,2.15,2.15,2.095,1.96,1.88,
1.701,1.570,1.439,1.327,1.127,0.927,0.927$
NACA-W-4-2412
$WGPLNF CHRDR=6.5,CHRDTP=4.5,SSPN=11.0,SSPNE=11.0,
SAVSI=11.0,CHSTAT=0.0,TWISTA=-3.0,DHDADI=3.0,TYPE=1.0$
NACA-H-5-23112
$HTPLNF CHRDR=3.0,CHRDTP=2.5,SSPN=4.0,SSPNE=4.0,
SAVSI=10.0,CHSTAT=0.00,TWISTA=0.0,DHDADI=8.00,TYPE=1.0$
NACA-V-4-0012
$VTPLNF CHRDR=3.583,CHRDTP=2.31837,SSPN=5.0,SSPNE=5.0,
SAVSI=25.0,CHSTAT=0.00,TYPE=1.0$
DAMP
NEXT CASE
```

b. DATCOM output file:

```

*****
*   USAF STABILITY AND CONTROL DIGITAL DATCOM   *
*   PROGRAM REV. JAN 96   DIRECT INQUIRIES TO:   *
*   WRIGHT LABORATORY (WL/FIGC)   ATTN: W. BLAKE   *
*   WRIGHT PATTERSON AFB, OHIO 45433   |
*   PHONE (513) 255-6764,   FAX (513) 258-4054   *
*****
1   CONERR - INPUT ERROR CHECKING
0   ERROR CODES - N* DENOTES THE NUMBER OF OCCURENCES OF EACH ERROR
0   A - UNKNOWN VARIABLE NAME
0   B - MISSING EQUAL SIGN FOLLOWING VARIABLE NAME
0   C - NON-ARRAY VARIABLE HAS AN ARRAY ELEMENT DESIGNATION - (N)
0   D - NON-ARRAY VARIABLE HAS MULTIPLE VALUES ASSIGNED
0   E - ASSIGNED VALUES EXCEED ARRAY DIMENSION
0   F - SYNTAX ERROR

0***** INPUT DATA CARDS *****

CASEID NAVION TEST
$FLTCON NMACH=1.0,MACH(1)=0.6$
$FLTCON NALT=1.0,ALT(1)=6500.0$
$FLTCON NALPHA=12.0,ALSCHD=-4.0,-2.0,0.0,2.0,4.0,6.0,8.0,
10.0,12.0,14.0,16.0,18.0,GAMMA=0.0,WT=4000.0,LOOP=1.0$
$OPTINS SREF=121.0,CBARR=5.56,BLREF=22.0$
$SYNTHS XCG=6.4,ZCG=-0.84,XH=4.0,ZW=-0.84,ALIW=2.0,XH=22.0,
ZH=0.0,ALIH=-3.0,XV=24.0,ZV=0.125,XVF=19.76,ZVF=1.25,VERTUP=.TRUE.$
$BODY NX=20.0,ITYPE=1.0,
X=0.0,0.958,1.9165,4.375,6.25,7.708,8.75,10.208,11.6665,13.125,14.583,
16.0415,17.5,18.958,20.41665,21.875,23.333,24.79165,26.25,27.708,
ZU=0.0,0.8835,1.1835,1.43750,1.76875,2.26040,2.5,2.2915,2.1875,1.9875,
1.8771,1.70625,1.659585,1.54854,1.4354,1.3415,1.209375,1.0098335,0.98229,0.871
ZL=-0.1,-1.271,-1.4875,-1.9146,-2.296,-2.296,-2.2915,-2.2915,-2.1875,-2.0835,
-1.9790,-1.875,-1.6665,-1.56250,-1.4585,-1.3540,-1.25,-1.146,-1.0415,-0.9375,
S=0.0,0.9765,1.177,1.3735,2.3145,2.2915,2.196,2.1005,2.005,1.8758,
1.65625,1.453605,1.38605,1.31720,1.24126,1.096,1.08434,0.853125,0.7520,0.51215
P=6.913,8.999,9.668,11.207,12.683,13.176,14.114,15.019,15.399,
15.533,15.003,14.765,13.749,11.702,9.299,7.039,5.618,2.292,
R=0.0,1.090,1.568,1.803,1.999,2.097,2.156,2.156,2.15,2.15,2.095,1.96,1.88,
1.701,1.570,1.439,1.327,1.127,0.927,0.927$
NACA-W-4-2412
$WGPLNF CHRDR=6.5,CHRDTP=4.5,SSPN=11.0,SSPNE=11.0,
SAVSI=11.0,CHSTAT=0.0,TWISTA=-3.0,DHDADI=3.0,TYPE=1.0$
NACA-H-5-23112
$HTPLNF CHRDR=3.0,CHRDTP=2.5,SSPN=4.0,SSPNE=4.0,
SAVSI=10.0,CHSTAT=0.00,TWISTA=0.0,DHDADI=0.00,TYPE=1.0$
NACA-V-4-0012
$VTPLNF CHRDR=3.583,CHRDTP=2.31837,SSPN=5.0,SSPNE=5.0,
SAVSI=25.0,CHSTAT=0.00,TYPE=1.0$
DAMP
NEXT CASE
1   THE FOLLOWING IS A LIST OF ALL INPUT CARDS FOR THIS CASE.
0
CASEID NAVION TEST
$FLTCON NMACH=1.0,MACH(1)=0.6$
$FLTCON NALT=1.0,ALT(1)=6500.0$
$FLTCON NALPHA=12.0,ALSCHD=-4.0,-2.0,0.0,2.0,4.0,6.0,8.0,
10.0,12.0,14.0,16.0,18.0,GAMMA=0.0,WT=4000.0,LOOP=1.0$
$OPTINS SREF=121.0,CBARR=5.56,BLREF=22.0$
$SYNTHS XCG=6.4,ZCG=-0.84,XH=4.0,ZW=-0.84,ALIW=2.0,XH=22.0,
ZH=0.0,ALIH=-3.0,XV=24.0,ZV=0.125,XVF=19.76,ZVF=1.25,VERTUP=.TRUE.$
$BODY NX=20.0,ITYPE=1.0,
X=0.0,0.958,1.9165,4.375,6.25,7.708,8.75,10.208,11.6665,13.125,14.583,
16.0415,17.5,18.958,20.41665,21.875,23.333,24.79165,26.25,27.708,
ZU=0.0,0.8835,1.1835,1.43750,1.76875,2.26040,2.5,2.2915,2.1875,1.9875,
1.8771,1.70625,1.659585,1.54854,1.4354,1.3415,1.209375,1.0098335,0.98229,0.871
ZL=-0.1,-1.271,-1.4875,-1.9146,-2.296,-2.296,-2.2915,-2.2915,-2.1875,-2.0835,
-1.9790,-1.875,-1.6665,-1.56250,-1.4585,-1.3540,-1.25,-1.146,-1.0415,-0.9375,
S=0.0,0.9765,1.177,1.3735,2.3145,2.2915,2.196,2.1005,2.005,1.8758,
1.65625,1.453605,1.38605,1.31720,1.24126,1.096,1.08434,0.853125,0.7520,0.51215
P=6.913,8.999,9.668,11.207,12.683,13.176,14.114,15.019,15.399,
15.533,15.003,14.765,13.749,11.702,9.299,7.039,5.618,2.292,
R=0.0,1.090,1.568,1.803,1.999,2.097,2.156,2.156,2.15,2.15,2.095,1.96,1.88,
1.701,1.570,1.439,1.327,1.127,0.927,0.927$
NACA-W-4-2412
$WGPLNF CHRDR=6.5,CHRDTP=4.5,SSPN=11.0,SSPNE=11.0,
SAVSI=11.0,CHSTAT=0.0,TWISTA=-3.0,DHDADI=3.0,TYPE=1.0$
NACA-H-5-23112
$HTPLNF CHRDR=3.0,CHRDTP=2.5,SSPN=4.0,SSPNE=4.0,
SAVSI=10.0,CHSTAT=0.00,TWISTA=0.0,DHDADI=0.00,TYPE=1.0$
NACA-V-4-0012
$VTPLNF CHRDR=3.583,CHRDTP=2.31837,SSPN=5.0,SSPNE=5.0,
SAVSI=25.0,CHSTAT=0.00,TYPE=1.0$
DAMP
NEXT CASE
0 INPUT DIMENSIONS ARE IN FT, SCALE FACTOR IS 1.0000

```

```

1          AUTOMATED STABILITY AND CONTROL METHODS PER APRIL 1976 VERSION OF DATCOM
            WING SECTION DEFINITION
0          IDEAL ANGLE OF ATTACK = 0.25757 DEG.

            ZERO LIFT ANGLE OF ATTACK = -1.87966 DEG.

            IDEAL LIFT COEFFICIENT = 0.25602
            ZERO LIFT PITCHING MOMENT COEFFICIENT = -0.05087
            MACH ZERO LIFT-CURVE-SLOPE = 0.09617 /DEG.

            LEADING EDGE RADIUS = 0.01587 FRACTION CHORD
            MAXIMUM AIRFOIL THICKNESS = 0.12000 FRACTION CHORD
            DELTA-Y = 3.16898 PERCENT CHORD

0          MACH= 0.6000 LIFT-CURVE-SLOPE = 0.11570 /DEG.      XAC = 0.26326
1          AUTOMATED STABILITY AND CONTROL METHODS PER APRIL 1976 VERSION OF DATCOM
            HORIZONTAL TAIL SECTION DEFINITION
0          IDEAL ANGLE OF ATTACK = 1.78972 DEG.

            ZERO LIFT ANGLE OF ATTACK = -1.09089 DEG.

            IDEAL LIFT COEFFICIENT = 0.32672
            ZERO LIFT PITCHING MOMENT COEFFICIENT = -0.01225
            MACH ZERO LIFT-CURVE-SLOPE = 0.09661 /DEG.

            LEADING EDGE RADIUS = 0.01587 FRACTION CHORD
            MAXIMUM AIRFOIL THICKNESS = 0.12000 FRACTION CHORD
            DELTA-Y = 3.16898 PERCENT CHORD

0          MACH= 0.6000 LIFT-CURVE-SLOPE = 0.11600 /DEG.      XAC = 0.26225
1          AUTOMATED STABILITY AND CONTROL METHODS PER APRIL 1976 VERSION OF DATCOM
            VERTICAL TAIL SECTION DEFINITION
0          IDEAL ANGLE OF ATTACK = 0.00000 DEG.

            ZERO LIFT ANGLE OF ATTACK = 0.00000 DEG.

            IDEAL LIFT COEFFICIENT = 0.00000
            ZERO LIFT PITCHING MOMENT COEFFICIENT = 0.00000
            MACH ZERO LIFT-CURVE-SLOPE = 0.09596 /DEG.

            LEADING EDGE RADIUS = 0.01587 FRACTION CHORD
            MAXIMUM AIRFOIL THICKNESS = 0.12000 FRACTION CHORD
            DELTA-Y = 3.16898 PERCENT CHORD

0          MACH= 0.6000 LIFT-CURVE-SLOPE = 0.11639 /DEG.      XAC = 0.26187
1          AUTOMATED STABILITY AND CONTROL METHODS PER APRIL 1976 VERSION OF DATCOM
            CHARACTERISTICS AT ANGLE OF ATTACK AND IN SIDESLIP
            WING-BODY-VERTICAL TAIL-HORIZONTAL TAIL CONFIGURATION

```

0	MACH= 0.6000 LIFT-CURVE-SLOPE = 0.11639 /DEG. XAC = 0.26187										
1	AUTOMATED STABILITY AND CONTROL METHODS PER APRIL 1976 VERSION OF DATCOM										
	CHARACTERISTICS AT ANGLE OF ATTACK AND IN SIDESLIP										
	WING-BODY-VERTICAL TAIL-HORIZONTAL TAIL CONFIGURATION										
	NAVION TEST										
	FLIGHT CONDITIONS						REFERENCE DIMENSIONS				
	MACH	ALTITUDE	VELOCITY	PRESSURE	TEMPERATURE	REYNOLDS	REF.	REFERENCE	LENGTH	MOMENT	REF. CENTER
	NUMBER	FT	FT/SEC	LB/FT**2	DEG R	NUMBER	AREA	LONG.	LAT.	HORIZ	VERT
						1/FT	FT**2	FT	FT	FT	FT
0	0.600	6500.00	654.66	1.6643E+03	495.497	3.5364E+06	121.000	5.560	22.000	6.400	-0.840
0	-----DERIVATIVE (PER DEGREE)-----										
0	ALPHA	CD	CL	CM	CN	CA	XCP	CLA	CMA	CYB	CNB
0											CLB
	-4.0	0.019	-0.168	0.1055	-0.169	0.008	-0.623	7.648E-02	-1.508E-02	-Inf	3.193E-03
	-2.0	0.016	-0.014	0.0793	-0.014	0.016	-5.481	7.772E-02	-1.055E-02		-1.371E-03
	0.0	0.017	0.143	0.0633	0.143	0.017	0.444	7.915E-02	-5.842E-03		-1.520E-03
	2.0	0.023	0.303	0.0560	0.303	0.012	0.185	8.109E-02	-5.119E-03		-1.674E-03
	4.0	0.033	0.467	0.0428	0.468	0.000	0.091	8.277E-02	-5.526E-03		-1.980E-03
	6.0	0.048	0.634	0.0339	0.635	-0.019	0.053	8.216E-02	-4.357E-03		-2.002E-03
	8.0	0.067	0.796	0.0254	0.797	-0.044	0.032	7.393E-02	-4.357E-03		-2.173E-03
	10.0	0.087	0.930	-0.0347	0.930	-0.076	-0.037	6.151E-02	-1.714E-02		-2.331E-03
	12.0	0.106	1.042	-0.1055	1.041	-0.113	-0.101	5.052E-02	-3.272E-02		-2.430E-03
	14.0	0.126	1.132	-0.1926	1.128	-0.152	-0.171	4.045E-02	-3.947E-02		-2.479E-03
	16.0	0.146	1.203	-0.2774	1.197	-0.192	-0.232	6.726E-03	-4.296E-02		-2.469E-03
	18.0	0.150	1.158	-0.3705	1.148	-0.216	-0.323	-5.164E-02	-4.447E-02		-2.414E-03
											-2.135E-03
0											
0		ALPHA	Q/QINF	EPSLON	D(EPSLON)/D(ALPHA)						
		-4.0	1.000	-0.779	0.375						
		-2.0	1.000	0.337	0.396						
		0.0	1.000	1.525	0.451						
		2.0	1.000	2.799	0.461						
		4.0	1.000	3.995	0.442						
		6.0	1.000	5.125	0.435						
		8.0	1.000	6.134	0.344						
		10.0	0.986	6.896	0.297						
		12.0	0.925	7.321	0.155						
		14.0	0.983	7.516	0.028						
		16.0	1.000	7.434	-0.143						
		18.0	1.000	6.946	-0.244						
1	AUTOMATED STABILITY AND CONTROL METHODS PER APRIL 1976 VERSION OF DATCOM										
	DYNAMIC DERIVATIVES										
	WING-BODY-VERTICAL TAIL-HORIZONTAL TAIL CONFIGURATION										
	NAVION TEST										

II) MATLAB CODES:

a. Multhopp Script by Meagan Shivers:

```
%This script was created by Meagan Shivers
%Made for AE 413
%This script contains the code for finding the coefficient of pitching
%moment of the fuselage of the Sig Rascal 72 using Multhopp's Theory
%The fuselage was split into 100 sections of upwash then 100 sections of
%downwash.
clear;
clc;
close all;
%Geometry of aircraft
%Fuselage length
lf = (27.708) - (6.5); %[ft]
lf_up = 4.375; %[ft]
lf_down = lf - lf_up; %[ft]
%Fuselage width
bf = [0.65,1.090,1.568,1.803,
2.15,2.15,2.095,1.96,1.88,1.701,1.570,1.439,1.327,1.127,0.927,0.927]; %[in]
%Fuselage width squared
```

```

bf_ftsq = (bf).^2; %[ft]
%Root and tip chord of wing
c_r = 6.5; %[ft]
c_t = 4; %[ft]
%Wing span
b = 22; %[ft]
%Height of tail from wing
h_h = 1.976;
%AC wing to AC tail
l_h = 16.043; %[ft]
%Sweep at leading edge
sweep_LE = 11;
sweep_quarter = 0;
%Finding the downwash model
%Finding taper ratio
lambda = c_t/c_r;
%Estimating wing surface area
s = (b/2)*c_r*(1+lambda); %[ft^2]
%Find MAC
c = (2/3)*c_r*((1+lambda+lambda^2)/(1+lambda)); %[ft]
%Find aspect ration
A = b^2/s;
%Find KA
K_A = (1/A) - (1/(1+A^1.7));
%Find K lambda
K_l = (10^-3*lambda)/7;
%Find KH
K_H = (1-(h_h/b))/((2*(l_h/b))^(1/3));
%Find downwash model
dE_da = 4.44*(K_A*K_l*K_H*(cosd(sweep_quarter))^(1/2))^1.19;
%For downwash section (after wing)
cm_a_down = 0;
%Split the fuselage behind the wing into sections of equal length
dx_down = lf_down/100;
%Create x1 for each of the 100 sections
x1_down = [];
for i = 1:100
    if i == 1
        x1_down(i) = 0.5*dx_down;
    else
        x1_down(i) = i*dx_down + 0.5*dx_down;
    end
end
dE_da_down = [];
%Create downwash model approximation
for k = 1:length(x1_down)
    dE_da_down(k) = (x1_down(k)/l_h)*(1-dE_da);

```

```

end
%Find down
for j = 1:length(xl_down)
%25 sections for bf(2)
if j > 1 && j <= 25
cm_a_down = bf_ftsq(2)*dE_da_down(j)*dx_down;
end
%25 sections for bf(3)
if j > 25 && j <= 50
cm_a_down = bf_ftsq(3)*dE_da_down(j)*dx_down;
end
%25 sections for bf(4)
if j > 50 && j <= 75
cm_a_down = bf_ftsq(4)*dE_da_down(j)*dx_down;
end
%25 sections for bf(5)
if j > 75 && j <= 100
cm_a_down = bf_ftsq(5)*dE_da_down(j)*dx_down;
end
end
%For the upwash section (before wing)
cm_a_up = 0;
%Split the fuselage in front of the wing into sections of equal length
dx_up = lf_up/100;
%Create xl for each of the 100 sections
xl_up = [];
for l = 1:100
if l == 1
xl_up(l) = 0.5*dx_up;
else
xl_up(l) = l*dx_up + 0.5*dx_up;
end
end
%Create and populate matrix holding the xl/cr for upwash
xl_cr = [];
for m = 1:length(xl_up)
xl_cr(m) = xl_up(m)/c_r;
end
%100 sections for upwash
dE_da_up = [];
for n = 1:length(xl_cr)
if n == length(xl_up)
dE_da_up(n) = 0.7756*(xl_cr(n)^-0.679);
end
dE_da_up(n) = (-0.179*log(xl_cr(n)))+0.2462;
end
%Find upwash contribution

```

```

for o = 1:length(dE_da_up)
cm_a_up = bf_ftsq(1)*(1+dE_da_up(o))*dx_up;
end
%Sum upwash and downwash to find total pitching moment of fuselage
cm_a_f = cm_a_up + cm_a_down;
fprintf('The pitching moment due to the downwash of the fuselage is: %f',cm_a_down);
fprintf('\nThe pitching moment due to the upwash of the fuselage is: %f',cm_a_up);
fprintf('\nThe total pitching moment due to the fuselage is: %f',cm_a_f);

```

b. Datcom Import Code:

```

close all
clear all
clc
% NAVION Example
alldata = datcomimport('C:\Users\vince\OneDrive\Desktop\413test\datcom.out', true,
0);
data = alldata{1}
% The missing data points will be filled with the values for the first alpha, since
these data points are meant to be used for all alpha values.
aerotab = {'cyb' 'cnb' 'clq' 'cmq'};
for k = 1:length(aerotab)
for m = 1:data.nmach
for h = 1:data.nalt
data.(aerotab{k})(:,m,h) = data.(aerotab{k})(1,m,h);
end
end
end
% Aerodynamic Coefficients
% You can now plot the aerodynamic coefficients:
%
% Plotting Lift Curve Moments
h1 = figure;
figtitle = {'Lift Curve' ''};
for k=1:1
subplot(2,1,k)
plot(data.alpha,permute(data.cl(:,k,:),[1 3 2]))
grid
ylabel(['Lift Coefficient (Mach =' num2str(data.mach(k)) ')'])
title(figtitle{k});
end
xlabel('Angle of Attack (deg)')
% Plotting Drag Polar Moments
h2 = figure;
figtitle = {'Drag Polar' ''};
for k=1:1
subplot(2,1,k)

```

```

plot(permute(data.cd(:,k,:),[1 3 2]),permute(data.cl(:,k,:),[1 3 2]))
grid
ylabel(['Lift Coefficient (Mach =' num2str(data.mach(k)) ' ')]
title(figtitle{k})
end
xlabel('Drag Coefficient')
% Plotting Pitching Moments
h3 = figure;
figtitle = {'Pitching Moment' ''};
for k=1:1
subplot(2,1,k)
plot(permute(data.cm(:,k,:),[1 3 2]),permute(data.cl(:,k,:),[1 3 2]))
grid
ylabel(['Lift Coefficient (Mach =' num2str(data.mach(k)) ' ')]
title(figtitle{k})
end
xlabel('Pitching Moment Coefficient')
alpha=[-4,-2,0,2,4,6,8,10,12,14,16,18];

```

c. 3D Plot Code by Dr. Greiner:

```

%*****
% The source code contained herein was developed for Embry-Riddle
% Aeronautical University by Glenn P. Greiner, Professor and Jafar
% Mohammed, Student Assistant of the Aerospace Engineering Department,
% Daytona Beach Campus. Copyright 2008. All rights reserved.
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% program is made available only for education and personal research. It
% may not be sold to other parties. If you copy some or all of the
% software you are requested to return a copy of any source additions that
% you believe make a significant improvement in its range of application.
%*****
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% datcom3d NAVION Input File %
% March 24, 2008 %
% Works with Version datcom3d v1.2 %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
clear
clc
clf
%% VISUALIZATION and RESOLUTION %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
wframe = 1; %0 = Shaded model
%1 = Wireframe model

```

```

fusres = 20; %Fuselage resolution
wgres = 20; %Wing,HT,VT resolution
%% (DO NOT CHANGE VALUES IN THIS BOX) %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
XW=0;ZW=0;ALIW=0;XH=0;ZH=0;ALIH=0;XV=0;YV=0;numVT=1;VERTUP=1; %%
NX=0;X=zeros(20);S=zeros(20);R=zeros(20);ZU=zeros(20);ZL=zeros(20); %%
CHRDRT_WG=0;CHRDRT_WG=0;CHRDTP_WG=0;SSPN_WG=0;SSPNOP_WG=0;SAVSI_WG=0; %%
SAVSO_WG=0;CHSTAT_WG=0;DHDADI_WG=0;DHDADO_WG=0;TC_WG=.12; %%
CHRDRT_HT=0;CHRDRT_HT=0;CHRDTP_HT=0;SSPN_HT=0;SSPNOP_HT=0;SAVSI_HT=0; %%
SAVSO_HT=0;CHSTAT_HT=0;DHDADI_HT=0;DHDADO_HT=0;TC_HT=.12; %%
CHRDRT_VT=0;CHRDRT_VT=0;CHRDTP_VT=0;SSPN_VT=0;SSPNOP_VT=0;SAVSI_VT=0; %%
SAVSO_VT=0;CHSTAT_VT=0;TC_VT=.12; %%
SPANFI_F=0;SPANFO_F=0;CHRDFT_F=0;CHRDFO_F=0;DELTA_F=0; %%
SPANFI_A=0;SPANFO_A=0;CHRDFT_A=0;CHRDFO_A=0;DELTA_A=0;DELTAR_A=0; %%
SPANFI_E=0;SPANFO_E=0;CHRDFT_E=0;CHRDFO_E=0;DELTA_E=0; %%
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%% INPUT PARAMETERS BELOW %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% SYNTHS parameters
XW = 4;
ZW = -0.84;
ALIW = 2.0;
XH = 22;
ZH = 0;
ALIH = -3.0;
XV = 24;
ZV = 0.125;
% BODY parameters
NX = 20;
X =
[0.0,0.958,1.9165,4.375,6.25,7.708,8.75,10.208,11.6665,13.125,14.583,16.0415,17.5,18.
958,20.41665,21.875,23.333,24.79165,26.25,27.708];
ZU =
[0.0,0.8835,1.1835,1.43750,1.76875,2.26040,2.5,2.2915,2.1875,1.9875,1.8771,1.70625,1.
659585,1.54854,1.4354,1.3415,1.209375,1.0098335,0.98229,0.871];
ZL = [-0.1,-1.271,-1.4875,-1.9146,-2.296,-2.296,-2.2915,-2.2915,-2.1875,-2.0835,-
1.9790,-1.875,-1.6665,-1.56250,-1.4585,-1.3540,-1.25,-1.146,-1.0415,-0.9375];
R =
[0.5,1.090,1.568,1.803,1.999,2.097,2.156,2.156,2.15,2.15,2.095,1.96,1.88,1.701,1.570,
1.439,1.327,1.127,0.927,0.927];
S = [0.0,0.9765,1.177,1.3735,2.3145,2.2915,2.196,2.1005,2.005,1.8758,
1.65625,1.453605,1.38605,1.31720,1.24126,1.096,1.08434,0.853125,0.7520,0.51215];
% WING parameters
CHRDTP_WG = 4;
SSPNE_WG = 15;
SSPN_WG = 15;
CHRDRT_WG = 6.5;
SAVSI_WG = 11;
CHSTAT_WG = 0.0;

```

```

TWISTA_WG = 0.0;
DHDADI_WG = 3.0;
DHDADO_WG = 0.0;
TC_WG = .10;
% HORIZONTAL TAIL parameters
CHRDTP_HT = 2.5;
SSPNE_HT = 4;
SSPN_HT = 4;
CHRRDR_HT = 3;
SAVSI_HT = 10.0;
CHSTAT_HT = 0.0;
TWISTA_HT = 0.0;
DHDADI_HT = 8.0;
DHDADO_HT = 0.0;
% VERTICAL TAIL parameters
CHRDTP_VT = 2.31837;
SSPNE_VT = 5;
SSPN_VT = 5;
CHRRDR_VT = 3.583;
SAVSI_VT = 25.0;
CHSTAT_VT = 0.0;
%%% PLOTTING %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
warning off MATLAB:divideByZero
hold on
plotFuselage(NX,X,S,R,ZU,ZL,fusres)
plotWing(XW,ZW,ALIW,CHRRDR_WG,CHRRBP_WG,CHRDTP_WG,SSPN_WG,SSPNOP_WG,SAVSI_WG,SAVSO_WG,
CHSTAT_WG,DHDADI_WG,DHDADO_WG,...
SPANFI_F,SPANFO_F,CHRDFI_F,CHRDFO_F,DELTA_F,SPANFI_A,SPANFO_A,CHRDFI_A,CHRDFO_A,DELTA
L_A,DELTAR_A,TC_WG,wgres)
plotHT(XH,ZH,ALIH,CHRRDR_HT,CHRRBP_HT,CHRDTP_HT,SSPN_HT,SSPNOP_HT,SAVSI_HT,SAVSO_HT,CH
STAT_HT,DHDADI_HT,DHDADO_HT,...
SPANFI_E,SPANFO_E,CHRDFI_E,CHRDFO_E,DELTA_E,TC_HT,wgres)
plotVT(XV,YV,ZV,CHRRDR_VT,CHRRBP_VT,CHRDTP_VT,SSPN_VT,SSPNOP_VT,SAVSI_VT,SAVSO_VT,CHST
AT_VT,VERTUP,TC_VT,wgres)
%%% VIEWPORT/FIGURE PROPERTIES %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
if wframe == 0
%colormap([1 .7 .1]) %Set a/c to gold
colormap([0 0 1]) %Set a/c to blue
shading interp %Interpolated shading
lighting gouraud %Smooth airplane mesh
%camlight right %Apply a light source
%Custom Lighting Options, Note:[X Y Z]
light('Position',[1 -2 1],'Style','infinite');
light('Position',[1 2 1],'Style','infinite');
light('Position',[0 0 -6],'Style','infinite');
else
colormap([1 1 1]) %Set a/c to white

```

```
end
axis off %Turn off axis
axis equal %Correct aspect ratio
camva(4.5) %Zoom in a/c to fit figure
view(3) %Apply initial viewport rotation
%camproj('perspective') %Perspective viewing (not R2006a compatible)
rotate3d on %Rotate icon enabled at start up
%showplottool('plotbrowser')
set(gcf, 'NumberTitle', 'off', 'Name', 'Aircraft Plot', 'Color', [1 1 1]);
```